



Liver Fibrosis:

Non-invasive tests and earlier diagnosis

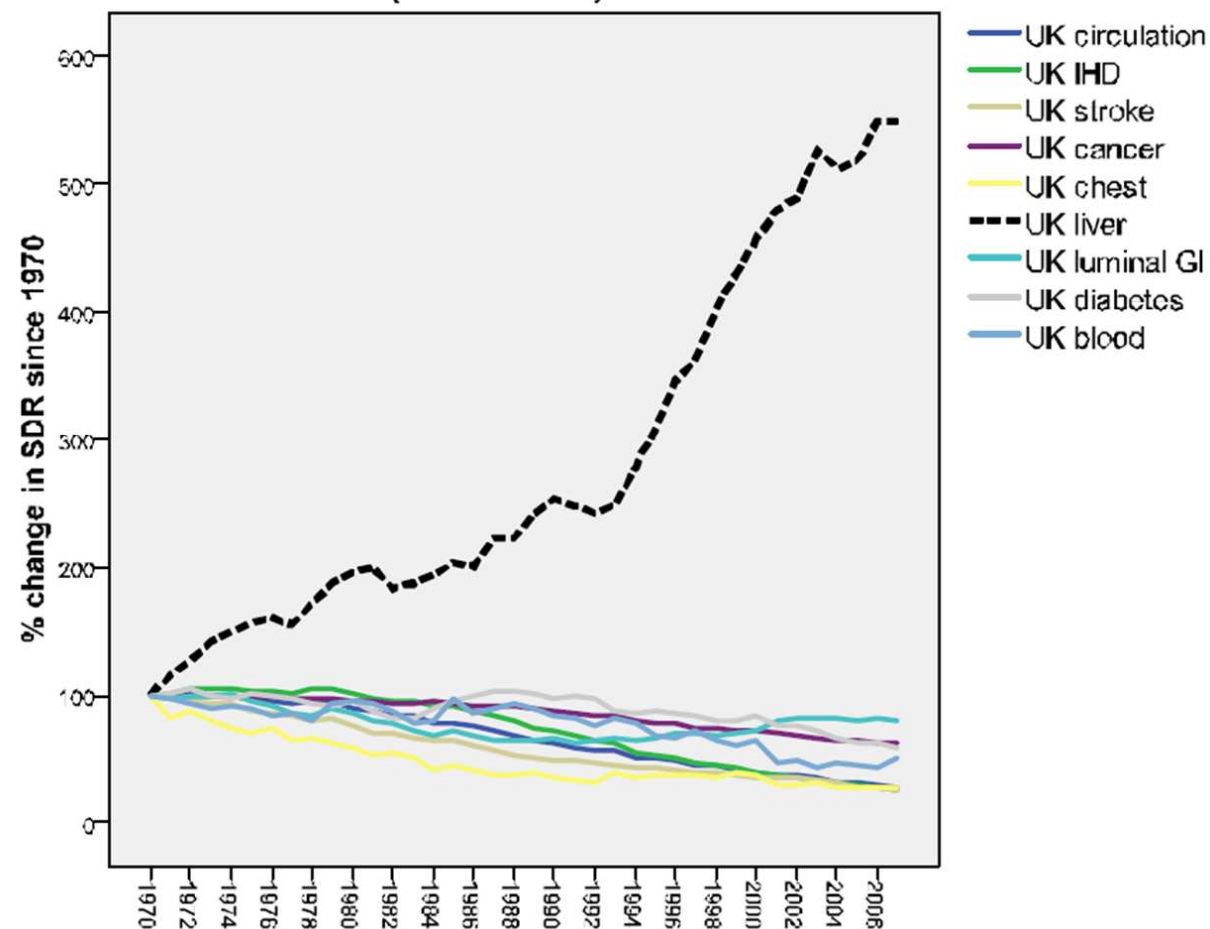
Dr Richard Aspinall

Consultant Hepatologist
Portsmouth Hospitals NHS Trust

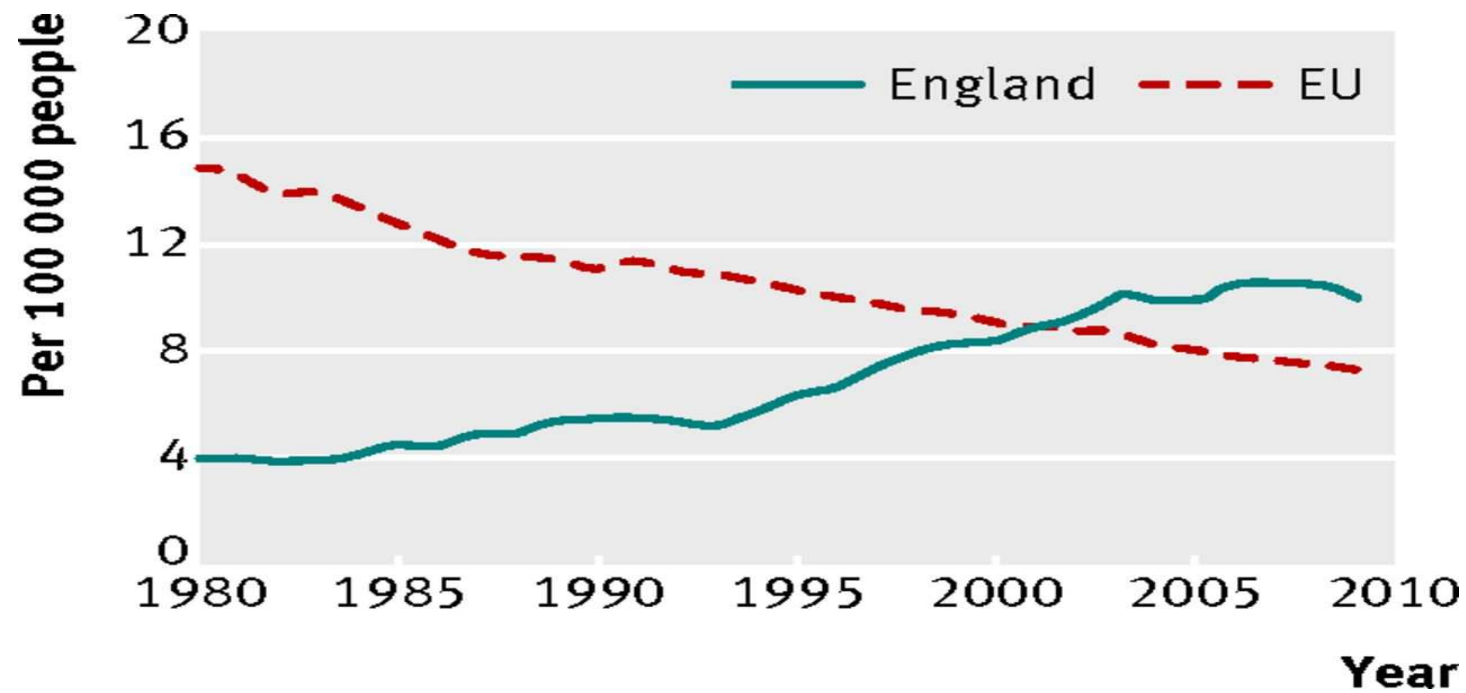
Increasing prevalence of chronic liver disease in the UK



UK under 65 standard death rate for various diseases
(1970 = 100%)



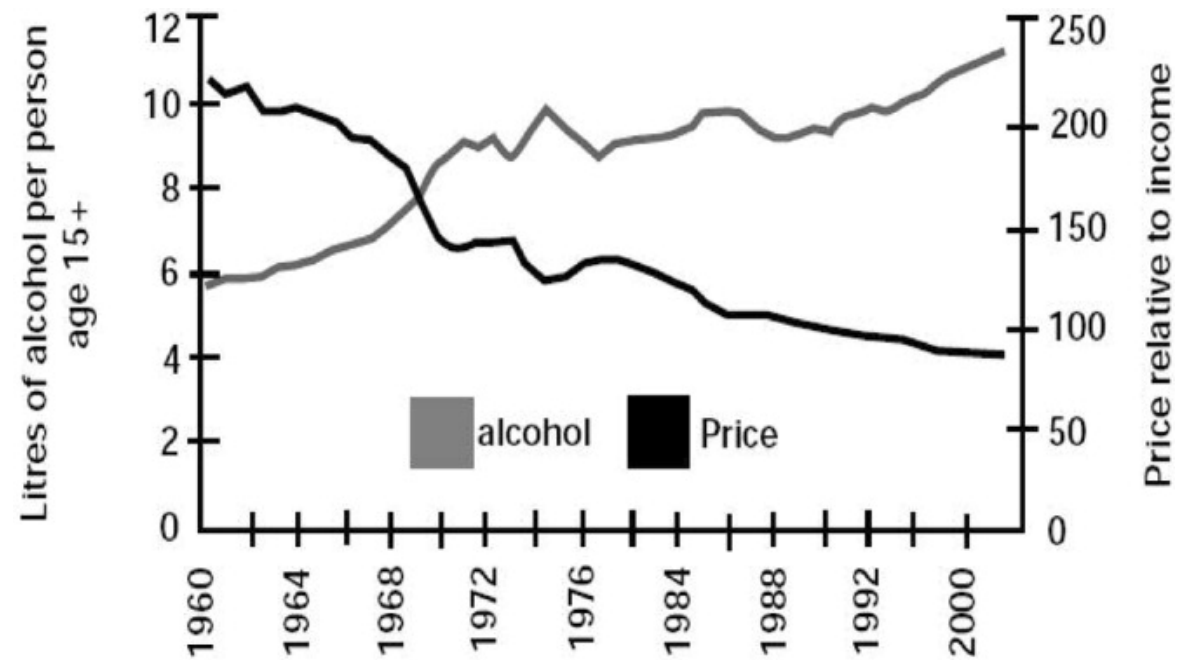
Deaths caused by liver disease in under 65s



Drivers of increased liver disease in UK

- Rising alcohol consumption (mainly)
- Patterns of alcohol consumption
- Obesity
- Viral hepatitis (HCV and others)

UK alcohol consumption



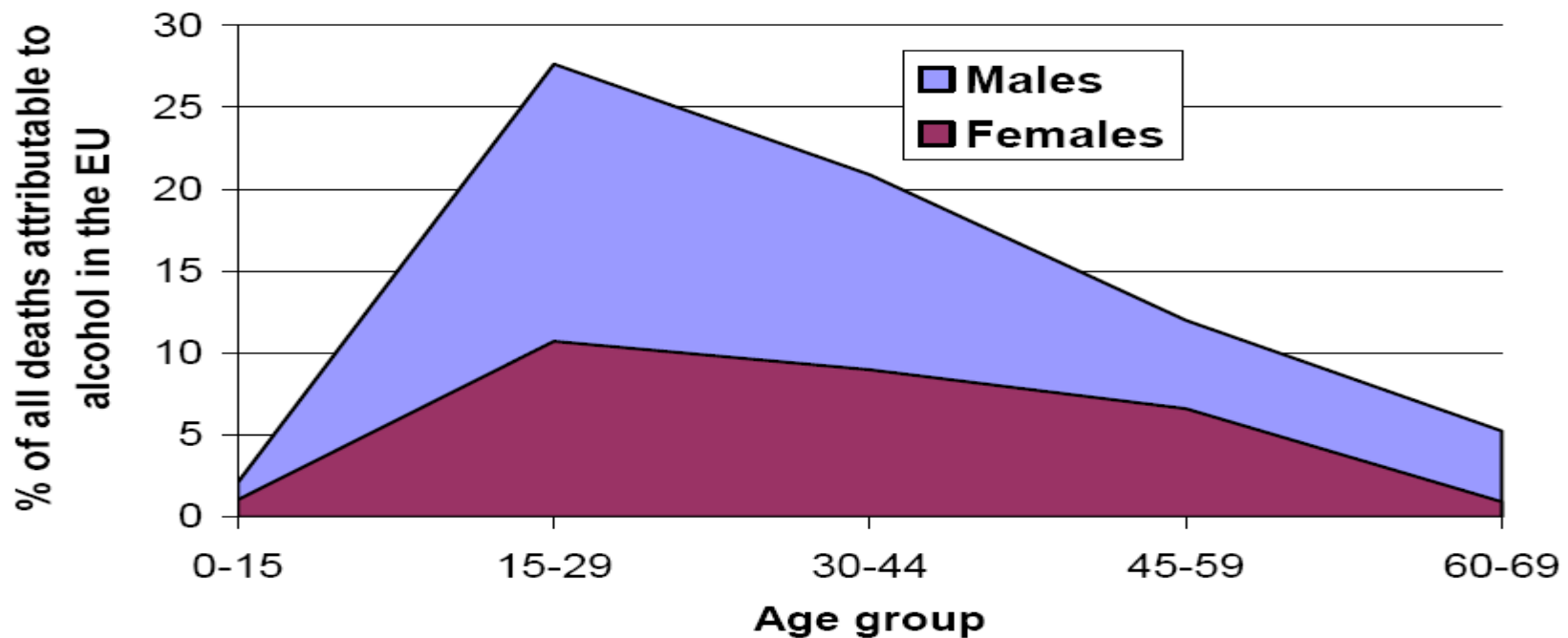
Data from Tighe, A. (ed.) *Statistical Handbook 2003*

£1.50/L



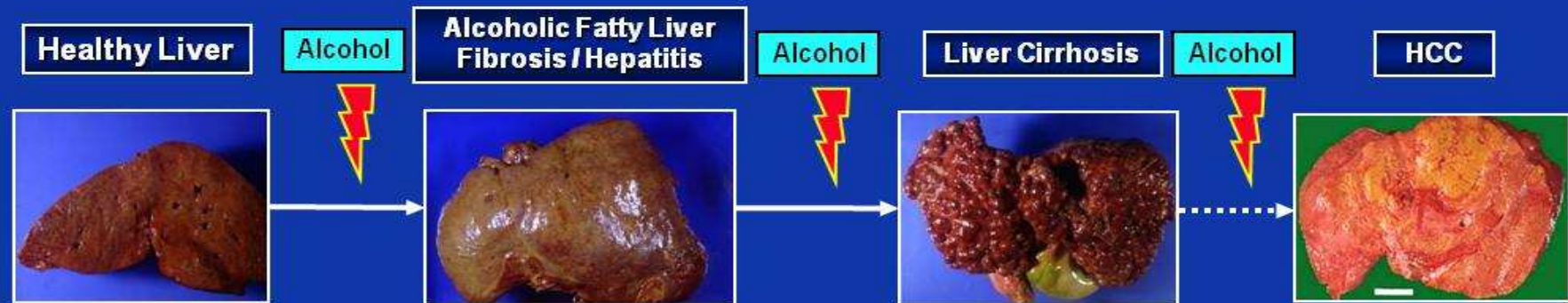
£0.85/L

Share of deaths attributable to alcohol in EU - disproportionately high among young people



Global burden of disease, Rehm, 2005

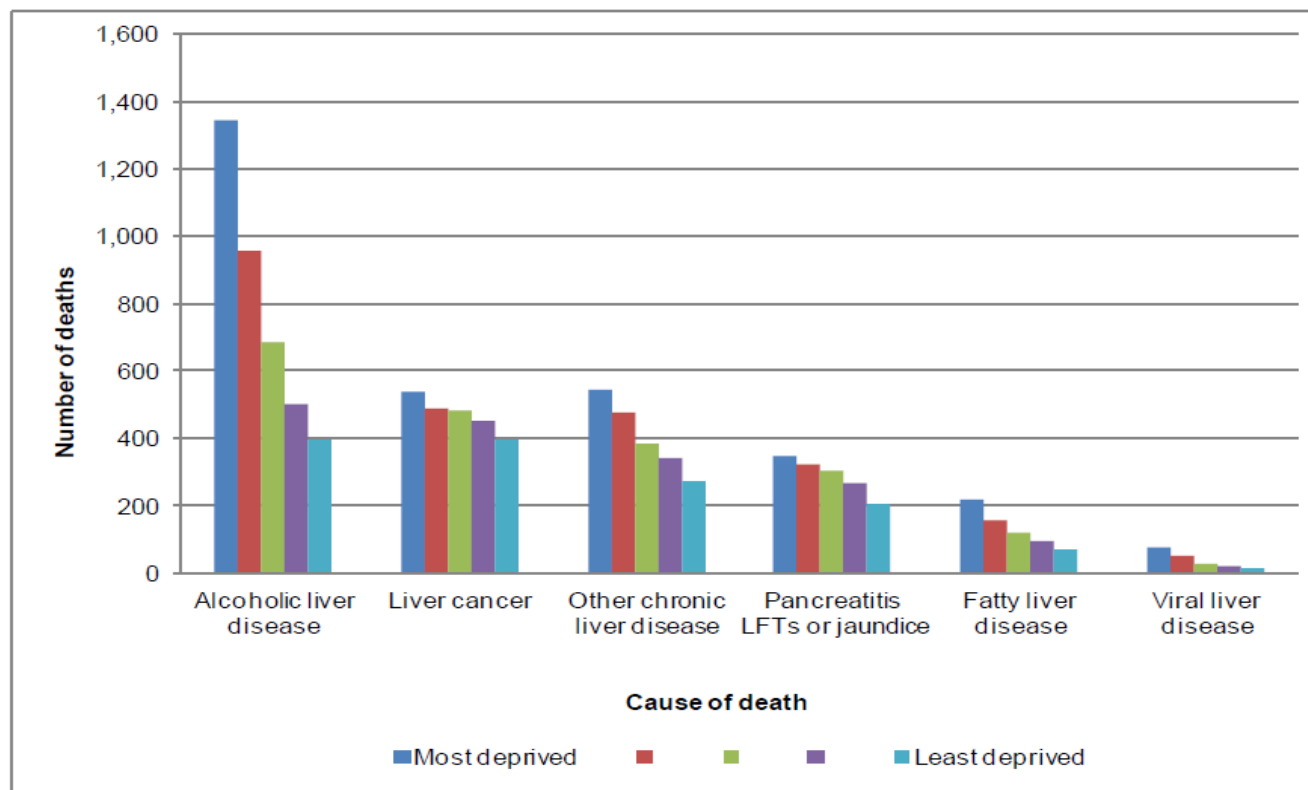
Alcoholic Liver Disease (ALD)



90-100% of alcoholics reveal steatosis
10-35% show signs of alcoholic fibrosis / hepatitis
10% develop cirrhosis

➡ Only a minority of heavy drinkers develop severe liver disease !

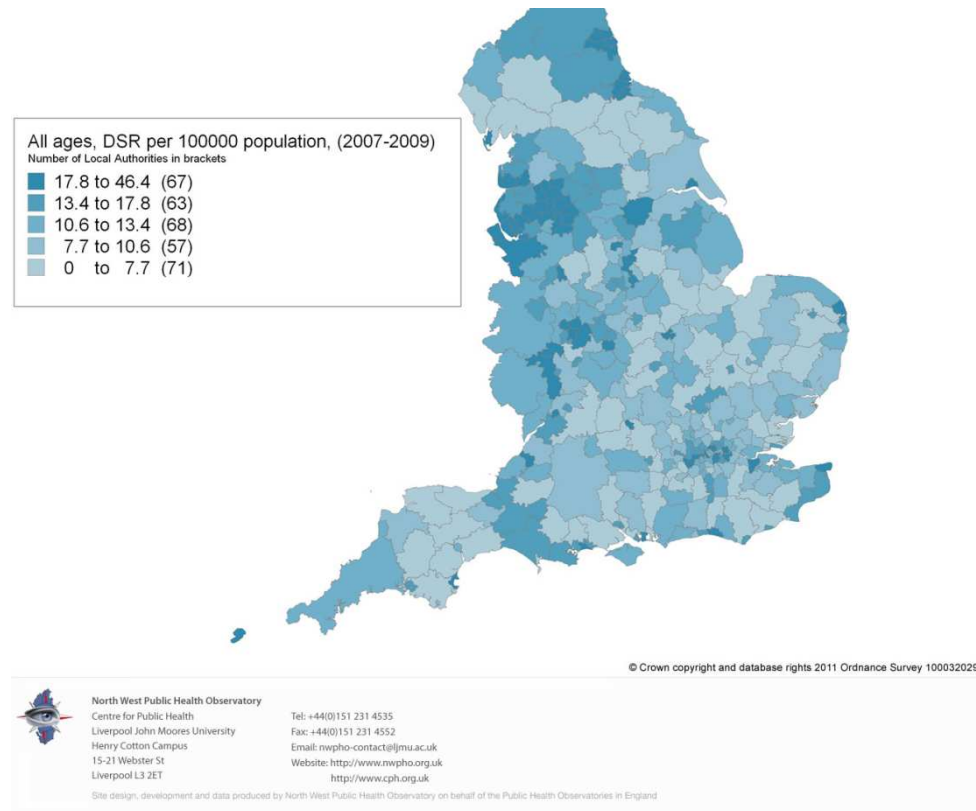
Social deprivation and deaths from liver disease



Note: IMD=Index of Multiple Deprivation

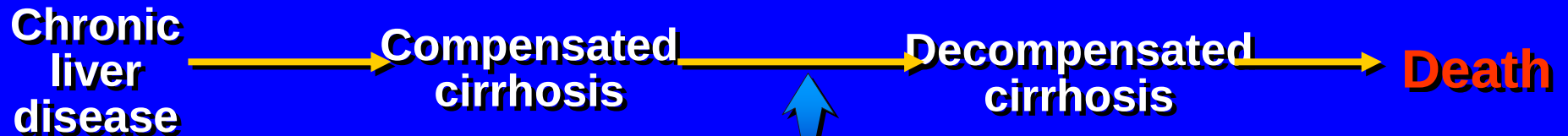
Source: ONS mortality data

Liver cirrhosis deaths



What is the natural history of cirrhosis?

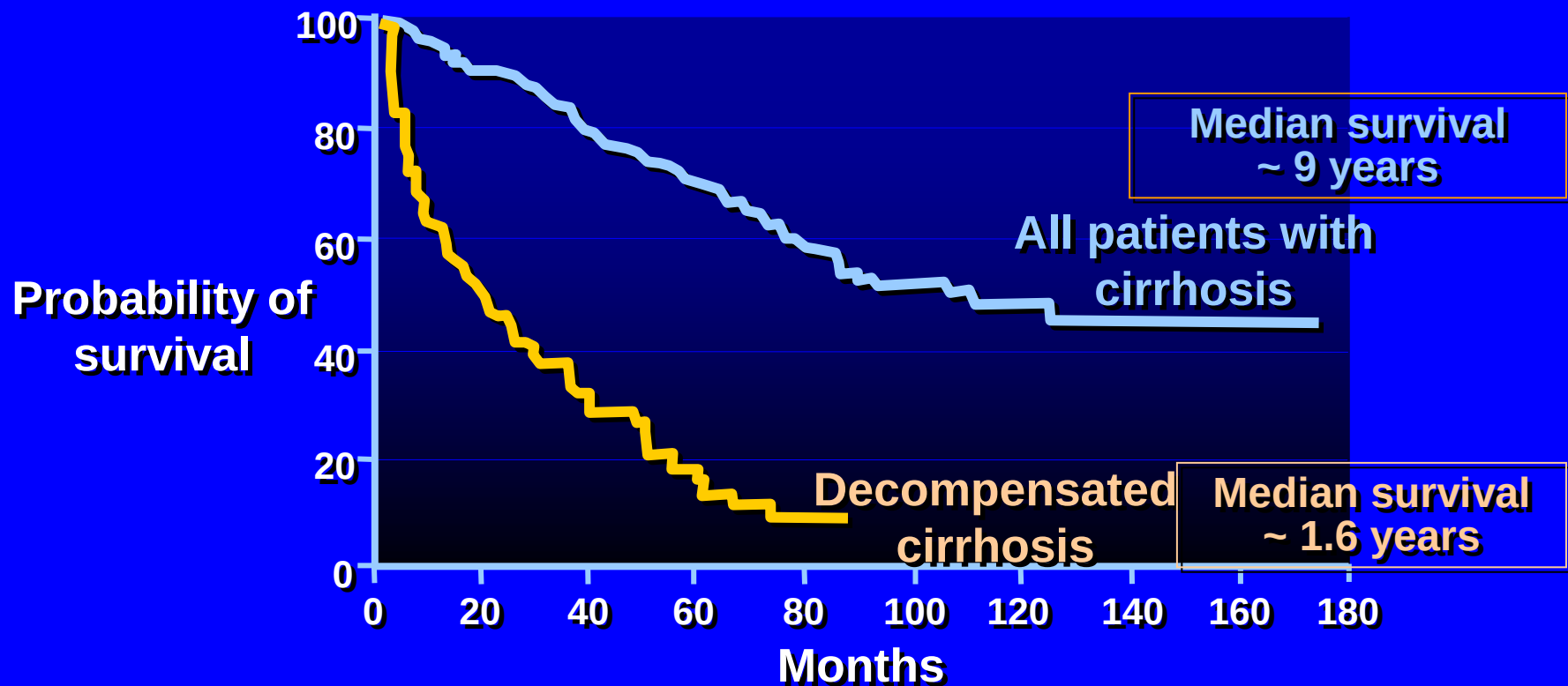
Natural History of Chronic Liver Disease



Development of complications:

- Variceal hemorrhage
- Ascites
- Encephalopathy
- Jaundice

Decompensation Shortens Survival



Gines et. al., Hepatology 1987;7:122

Risk of death in cirrhosis

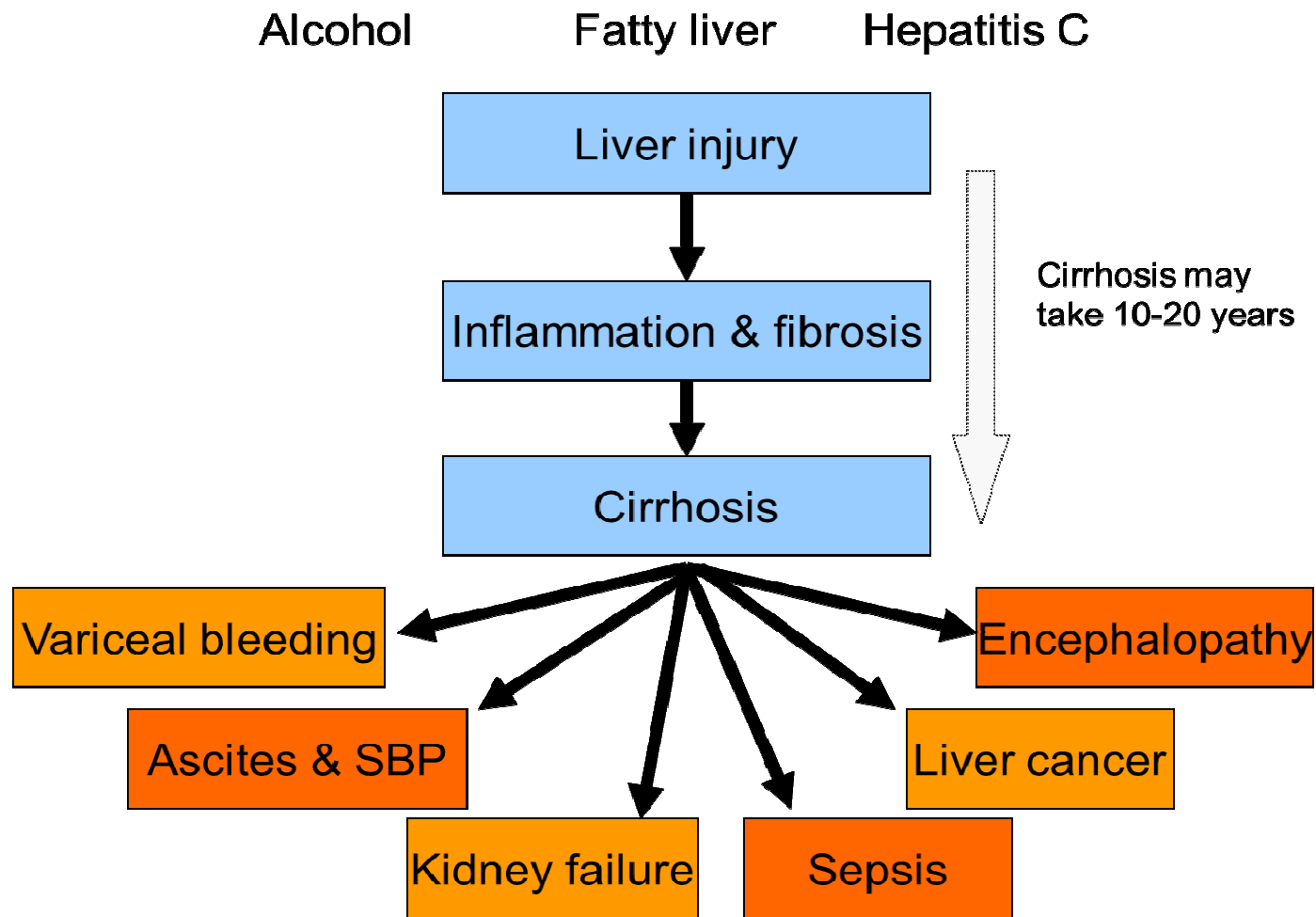
Medscape® www.medscape.com			
Clinical Status	Definition	Cumulative Probability of Death Per 1 Year	Cumulative Probability of Exiting this Status Per 1 Year
Compensated cirrhosis [†]			
• Stage 1	No varices, no ascites	1%	11%
• Stage 2	Varices, no ascites	3.4%	10%
Decompensated cirrhosis [†]			
• Stage 3	Ascites ± varices	20%	7.6%
• Stage 4	Bleeding ± ascites	57%	—

*From D'Amico G, Garcia-Tsao G, Pagliaro L. Natural history and prognostic indicators of survival in cirrhosis: a systematic review of 118 studies. J Hepatol 2006;44:217–231; and de Franchis R. Evolving consensus in portal hypertension: report of the Baveno IV consensus workshop on methodology of diagnosis and therapy in portal hypertension. J Hepatol 2005;43:167–176.⁹⁵

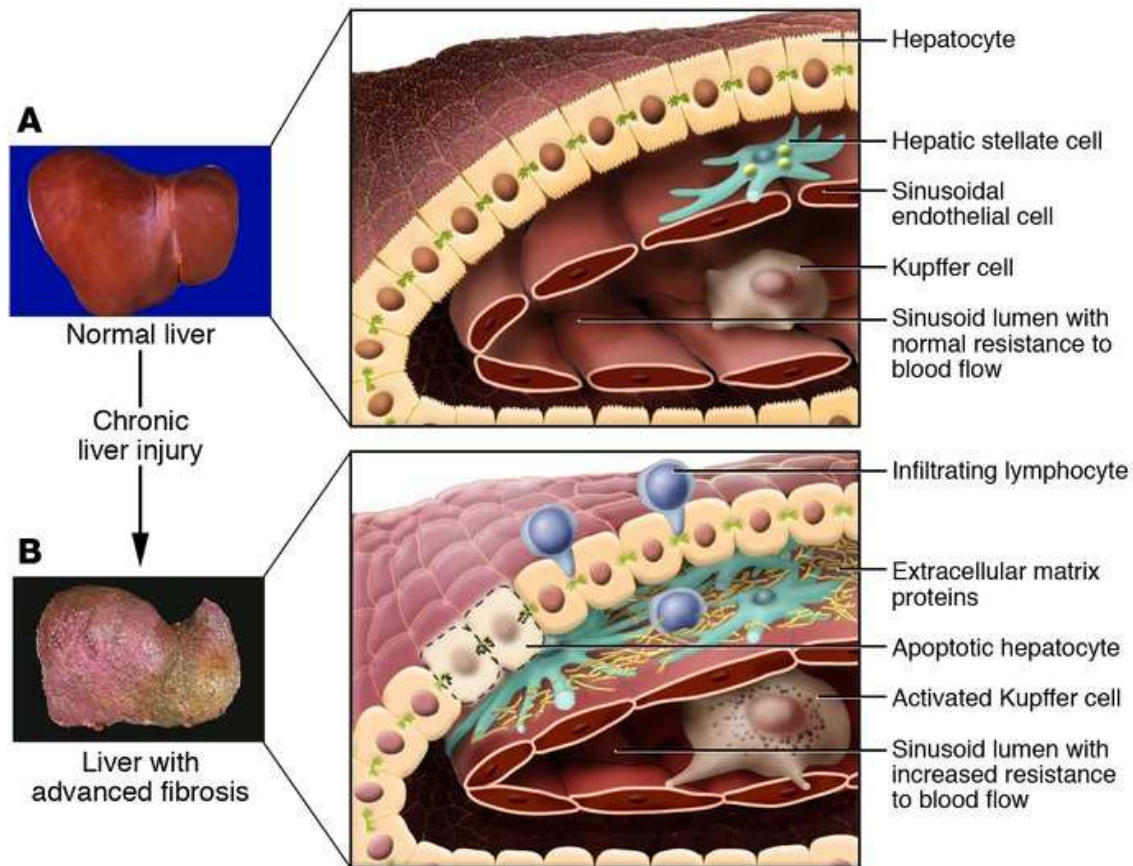
[†]Decompensated cirrhosis is defined by the presence of ascites, variceal bleeding, and/or encephalopathy.

Source: Semin Liver Dis © 2008 Thieme Medical Publishers

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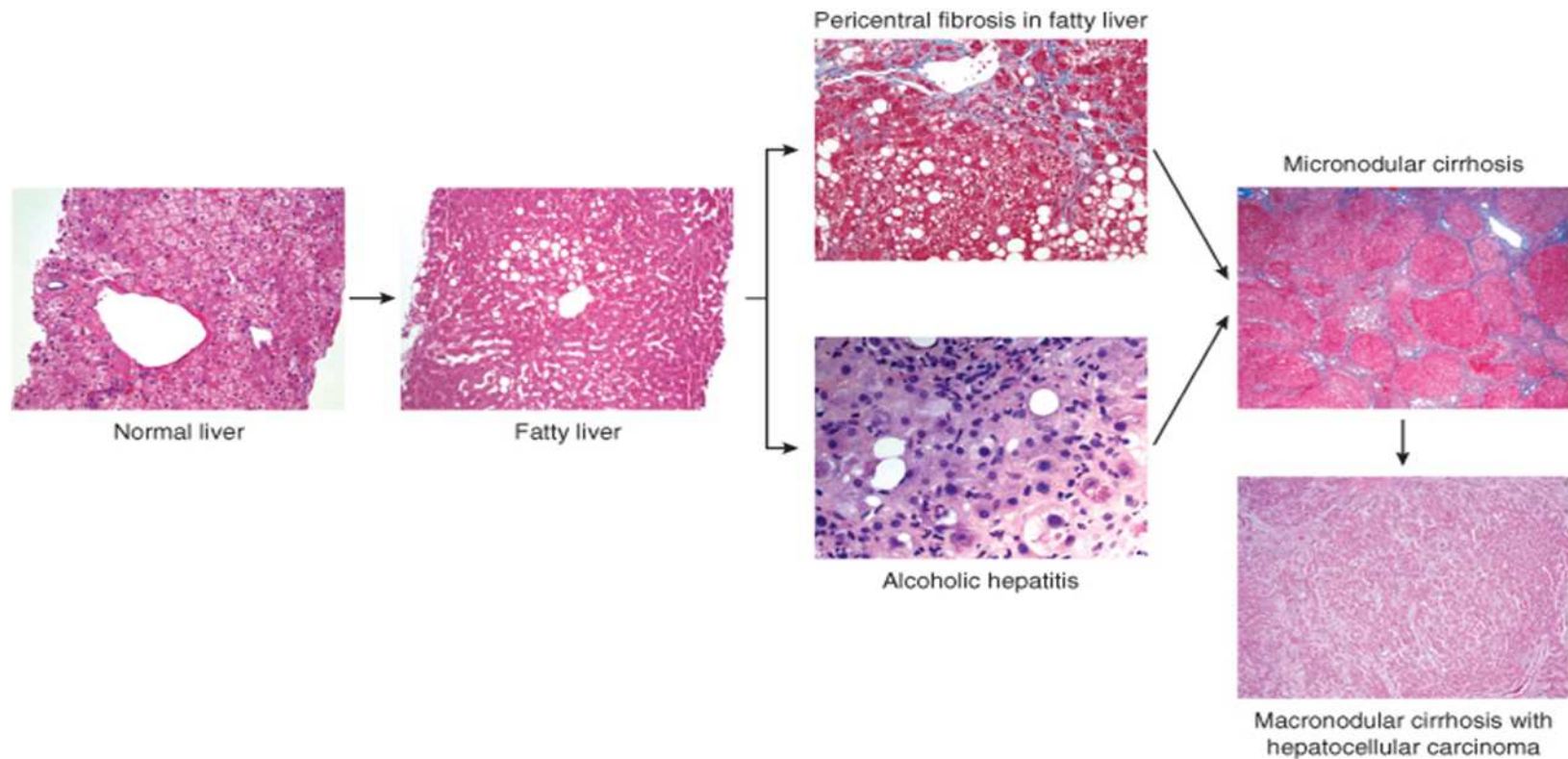


What is Liver Fibrosis?



Bataller R, Brenner DA. *J Clin Invest* 2005

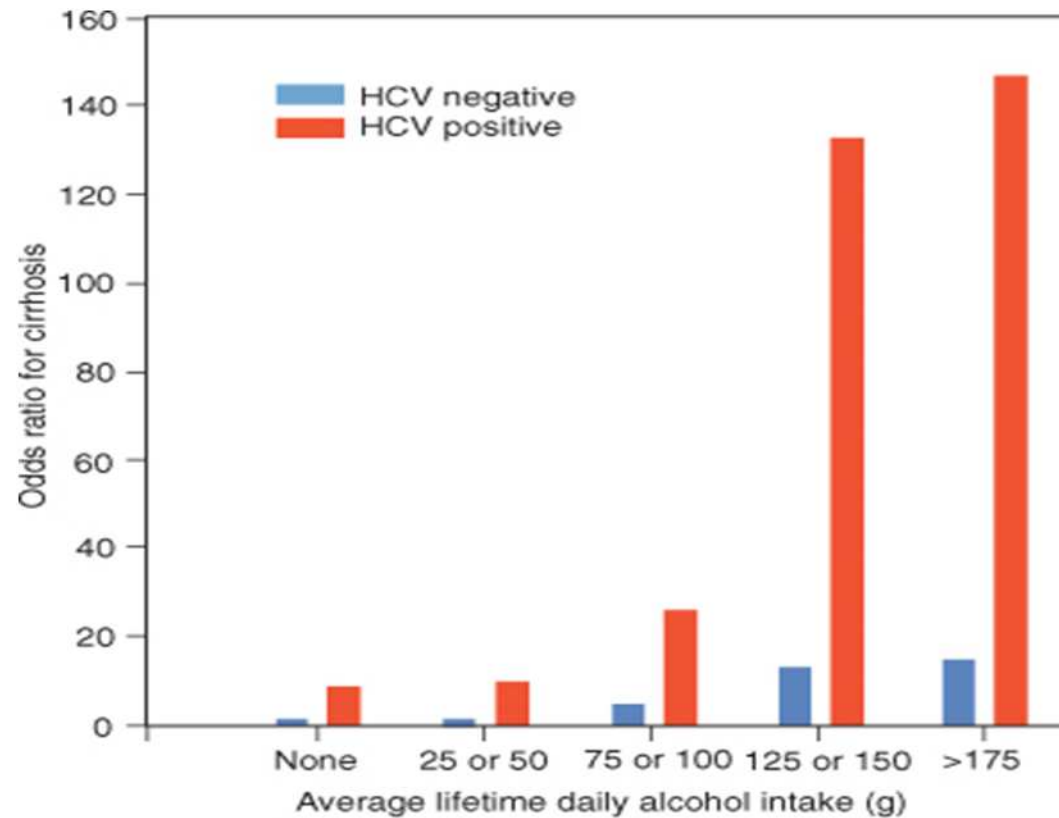
Spectrum of alcohol-induced liver injury



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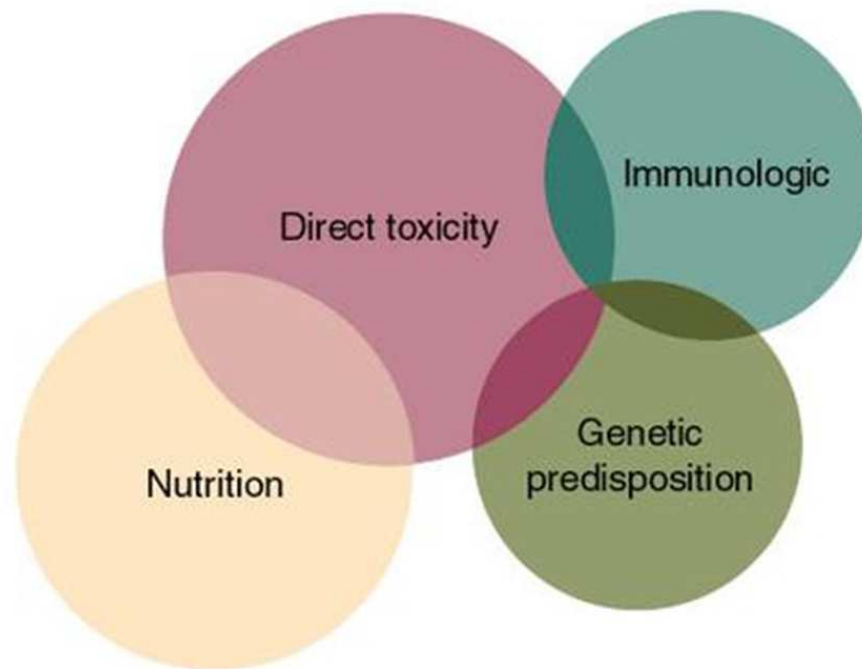
Alcohol + Hepatitis C = More cirrhosis



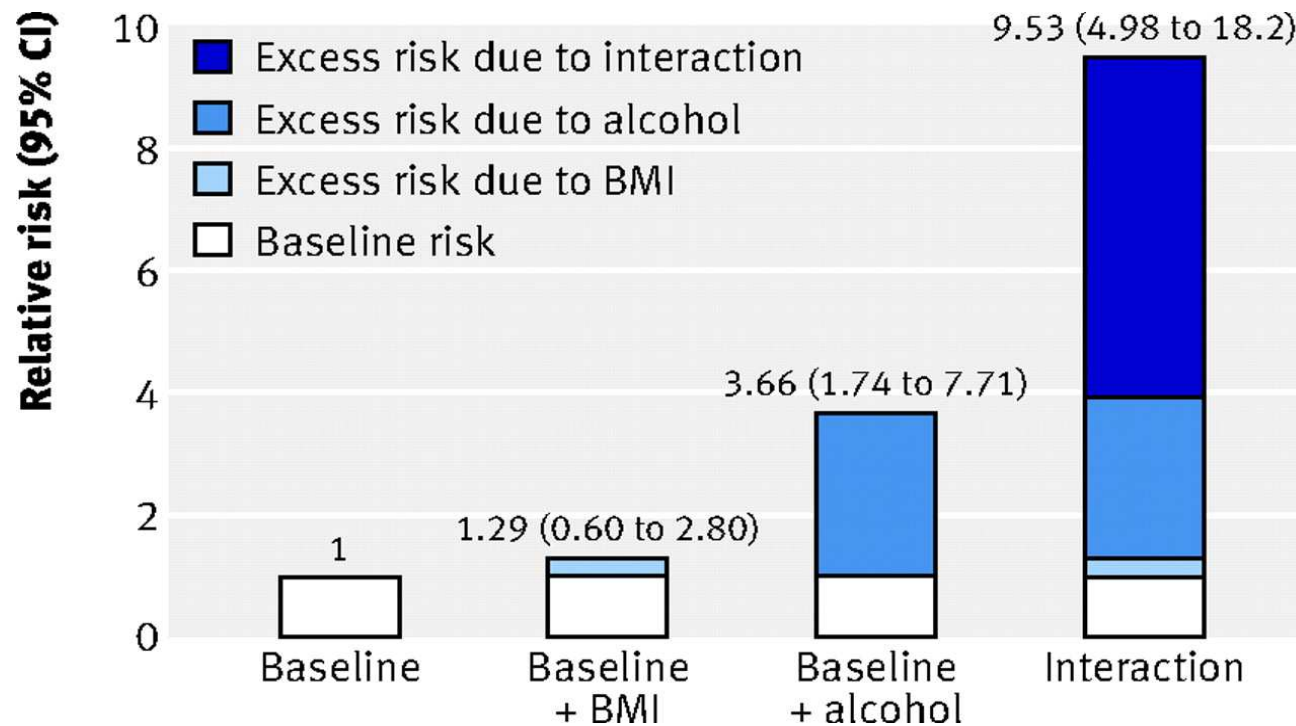
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Corrao G, et al: *Eur J Epidemiol* 1994; 10:657

Mechanism of liver damage due to alcohol



Influence of obesity and alcohol intake on deaths from liver disease in Scotland



Alcohol > 15 units / week BMI > 30
Scottish Midspan cohort study of 9559 men from 1965-1987

Hart CL et al. *BMJ* 2010

Major lifestyle determinants of fatty liver

What we eat



Excess Calories
Fat/ saturated fat
Sugars
Low fiber

What we don't do enough of



Sedentary lifestyle
Physical activity as treatment



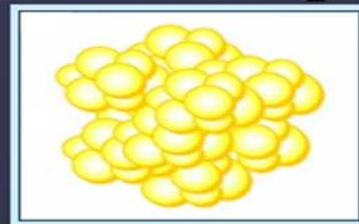
What we drink

Excess
Sucrose
Fructose
HFCS
Caramel

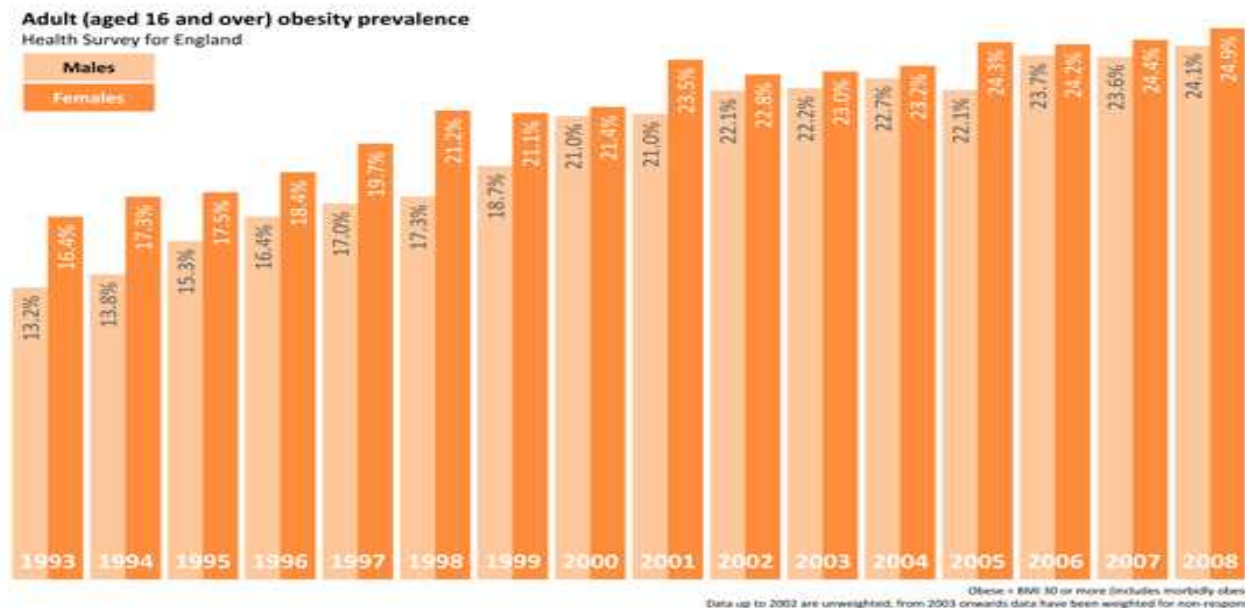


What we have in plenty

Obesity
Diets & drugs
for weight reduction



Rising obesity in England

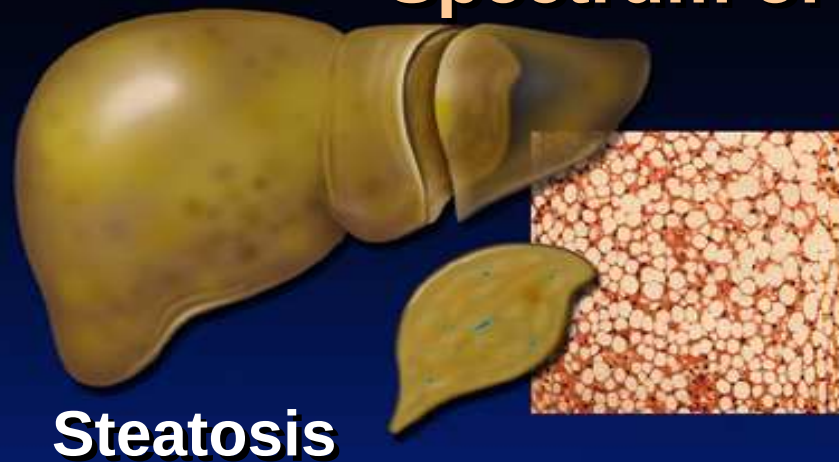


- By 2050, the national prevalence of obesity is predicted to affect 60% of adult men, 50% of adult women and 25% of children
- In Portsmouth, 24% of 10 year olds are already obese

Data sources: (1) Health Survey for England 2008, Department of Health; (2) Portsmouth City Council Healthy Weight Strategy 2011

Non Alcoholic Fatty Liver Disease (NAFLD)

Spectrum of Liver Pathology



Hepatocellular carcinoma



Epidemiology of NAFLD

- Commonest cause of abnormal LFTs
- Present in 20-30% of Western adults
- NASH is less common ~ 2-3% of general population
- If BMI >35, NAFLD = 91%, NASH = 37%
- Type 2 Diabetes: NAFLD = 70%
- Patients with T2DM are 4x more likely to have abnormal ALT than matched controls

Skelly et al. *J Hepatol.* 2001; 35: 195-199
Machado et al. *J Hepatol.* 2006; 45: 600-606
West et al. *QJM.* 2006; 99:871-876

Bedogoni et al. *Hepatology* 2005; 42: 44-52
Targher et al. *Diabetes Care.* 2006; 29: 1325-1330

How can we detect liver fibrosis?

Current options for fibrosis assessment



Liver Biopsy



**Serum
Biomarkers**



Imaging



Elastography

What does a liver biopsy add?

- **Diagnosis** – cannot distinguish ALD from NAFLD/NASH
- **Staging** – eg fibrosis versus cirrhosis
- **Prognostic information**
 - Functional tests are better – Child's-Pugh, MELD, HVPG, etc
- Diagnosis of **cirrhosis** to implement screening for HCC, varices, etc
 - Probably major indication in 2012

Is liver biopsy really the “gold standard” for staging liver fibrosis?

- Only 1 / 50,000 of liver volume sampled
- Sampling variation
- Right lobe versus left lobe
- Some inter-observer variation between pathologists and scoring systems
- Not without risk

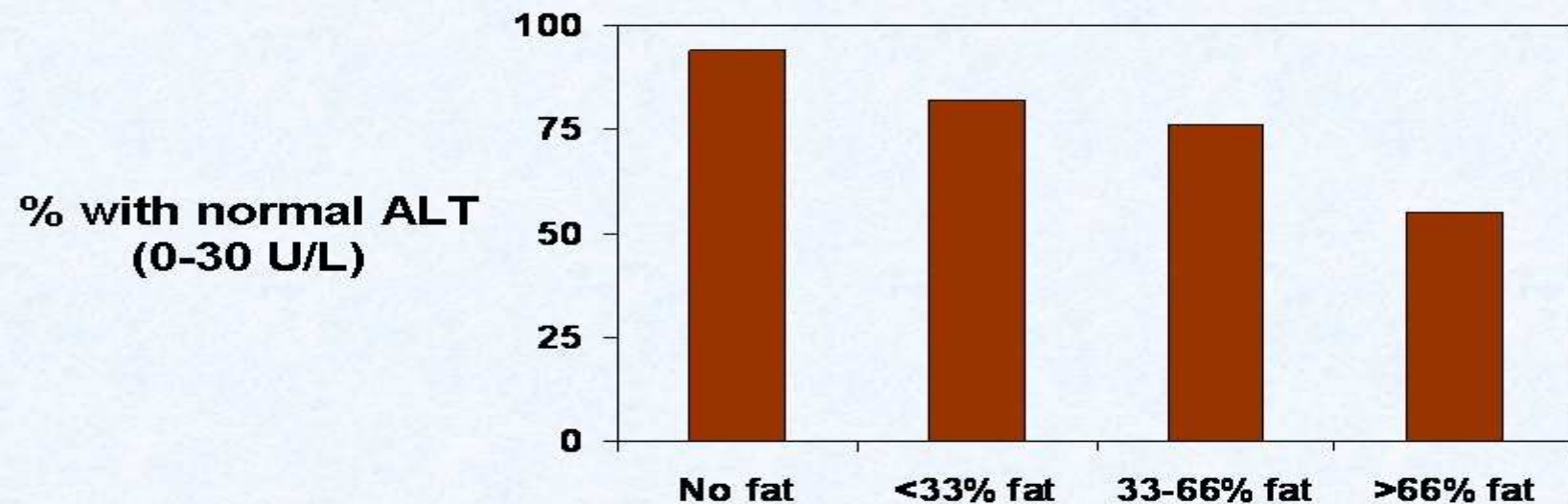
“Routine” blood tests:
How useful is ALT?

Is there a reliable cut-off ALT value?

- Short answer...no! (unfortunately)
- ALFIE project (Abnormal Liver Function In primary careE), funded by NIHR/HTA, Tayside
- Big primary care-based study - 15 years (approximately 2.3 million tests in 95,000 people)
- **Cut-offs are arbitrary - it is important to treat the LFT results as continuous variables**

Poor sensitivity of ALT for NAFLD

- 528 bariatric surgery patients (Belgium)



Luyckx et al, Int J Obes Relat Metab Disord (1998) 22: 222-226

“Routine tests” - particular concerns or alarm features

- Low platelet count
 - Suggests advanced fibrosis / cirrhosis
- Impaired synthetic liver function
 - Hypoalbuminaemia
 - INR / PT
- Low sodium level; kidney dysfunction

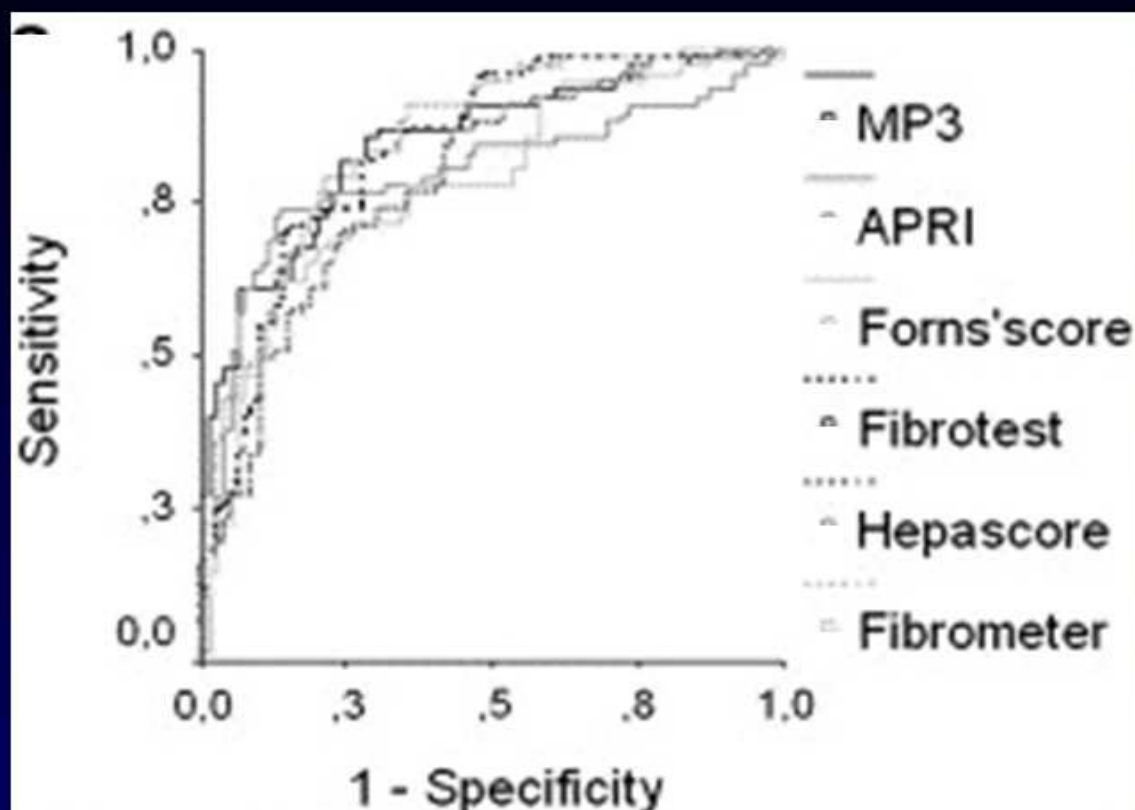
What about serum markers of liver fibrosis?

- There's a range of tests out there
- Not much between them
- Some are based on common laboratory and clinical parameters (i.e. cheap)
- Likely to be used more in future for screening of abnormal LFTs (eg NAFLD)

Serum fibrosis markers

- 1. Direct tests: markers related to the pathophysiology of fibrosis (components from the scar matrix)**
- 2. Indirect tests: parameters related to liver function and metabolism (ALT, Bili, INR, Alb, platelets.....)**
- 3. Tests combining 1 and 2**

Comparison of 6 biomarker scores in 180 pts with CHC (F0-1 vs. F2-4)



All with AUROCs 0.80-0.85 = only modest tests to distinguish no/mild vs. significant fibrosis

MP3: PIIINP, MMP-1

APRI: AST / platelets

Forns: γ GT, cholesterol, platelets, age

Fibrotest: age, Bili, γ GT, γ -globulin, haptoglobin, α 2M

Hepascore: age, gender, α 2M, HA, Bili

Fibrometer: age, INR, platelets, AST, urea, α 2M, HA

Leroy V et al, J Hepatol 2007

Simple scoring systems can identify advanced fibrosis...

Table 3 A comparison of the performance of each test for the diagnosis of advanced fibrosis in 145 patients with non-alcoholic fatty liver disease (NAFLD)

Test	AUROC (95% CI)	Cut-off	Sens (%)	Spec (%)	PPV (%)	NPV (%)
AST/ALT ratio	0.83 (0.74 to 0.91)	0.8	74	78	44	93
		1	52	90	55	89
APRI	0.67 (0.54 to 0.8)	1	27	89	37	84
BARD score	0.77 (0.68 to 0.87)	2	89	44	27	95
FIB-4 score	0.86 (0.78 to 0.94)	1.30	85	65	36	95
		3.25	26	98	75	85
NAFLD fibrosis score	0.81 (0.71 to 0.91)	-1.455	78	58	30	92
		0.676	33	98	79	86

ALT, alanine aminotransferase; APRI, AST-to-platelet ratio index; AST, aspartate aminotransferase; AUROC, area under the receiver operating characteristic curve; PPV, positive predictive value; NPV, negative predictive value; Sens, sensitivity; Spec, specificity.

Scores based on:

- Age
- Body Mass Index
- Diabetes Y/N?
- AST / ALT
- Platelets
- Albumin

...and may help avoid liver biopsy

	Cut-off	Patients avoiding liver biopsy*	False negative result
AST/ALT	<0.8	100/145 (69%)	7 (7%)
BARD score	<2	55/145 (38%)	3 (5%)
FIB-4 score	<1.30	90/145 (62%)	4 (5%)
NAFLD fibrosis score	<-1.455	75/145 (52%)	6 (8%)

*Patients with a value below the cut-off.

ALT, alanine aminotransferase; AST, aspartate aminotransferase; NAFLD, non-alcoholic fatty liver disease.

NAFLD fibrosis score calculator - Microsoft Internet Explorer provided by IPHIS

http://nafldscore.com/index.php

File Edit View Favorites Tools Help

NAFLD fibrosis score calculator

Page Tools

NAFLD fibrosis score

Online calculator

Angulo P, Hui JM, Marchesini G et al. **The NAFLD fibrosis score**
A noninvasive system that identifies liver fibrosis in patients with NAFLD
Hepatology 2007;45(4):846-854 [doi:10.1002/hep.21496](https://doi.org/10.1002/hep.21496)

Age (years)

BMI (kg/m²)

IGF/diabetes ☐

AST

ALT

Platelets (x10⁹/l)

Albumin (g/l)

BMI: body mass index
IGF: impaired fasting glucose

© 2009 nafldscore.com

concept: Dr Matthew Armstrong
site construction and design: Dr Jeremy Jones

Internet 100%

<http://nafldscore.com>

NAFLD fibrosis score calculator - Microsoft Internet Explorer provided by IPHIS

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NAFLD fibrosis score calculator

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Age (years)

BMI (kg/m²)

IGF/diabetes ☒

AST

ALT

Platelets (x10⁹/l)

Albumin (g/l)

Score

Original score

< -1.455: predictor of **absence** of significant fibrosis (F0-F2 fibrosis)
≤ -1.455 to ≤ 0.675: indeterminate score
> 0.675: predictor of **presence** of significant fibrosis (F3-F4 fibrosis)

BMI: body mass index
IGF: impaired fasting glucose

© 2009 naflscore.com

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Age (years)

BMI (kg/m²)

IGF/diabetes ☐

AST

ALT

Platelets (x10⁹/l)

Albumin (g/l)

Score -0.097

< -1.455: predictor of **absence** of significant fibrosis (F0-F2 fibrosis)
≤ -1.455 to ≤ 0.675: indeterminate score
> 0.675: predictor of **presence** of significant fibrosis (F3-F4 fibrosis)

BMI: body mass index
IGF: impaired fasting glucose

© 2009 naflscore.com

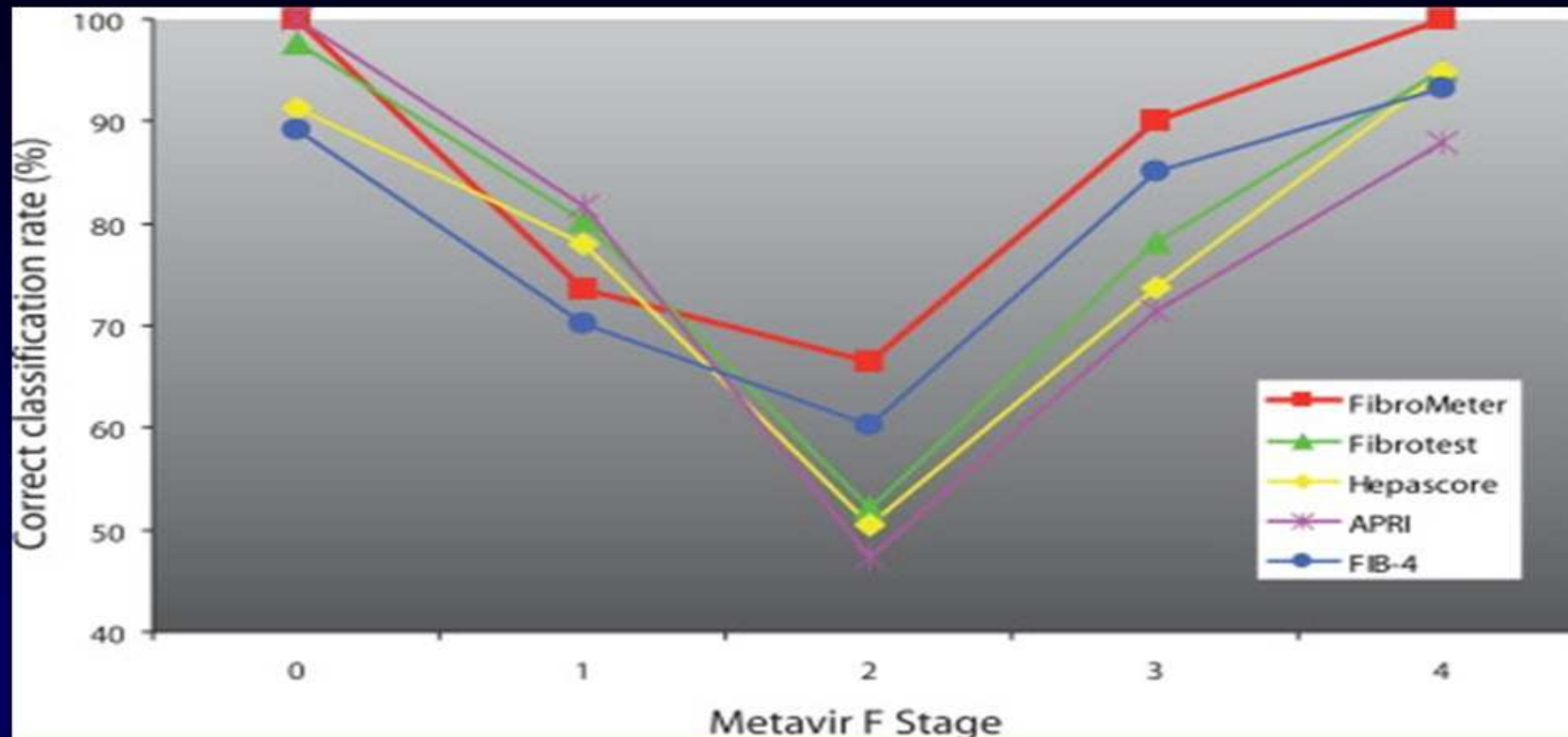
concept: Dr Matthew Armstrong
site construction and design: Dr Jeremy Jones

Done Internet 100%

<http://naflscore.com/>

Failure to correctly classify intermediate fibrosis stages

1056 patients with chronic hepatitis C; biopsy as (tarnished) gold standard

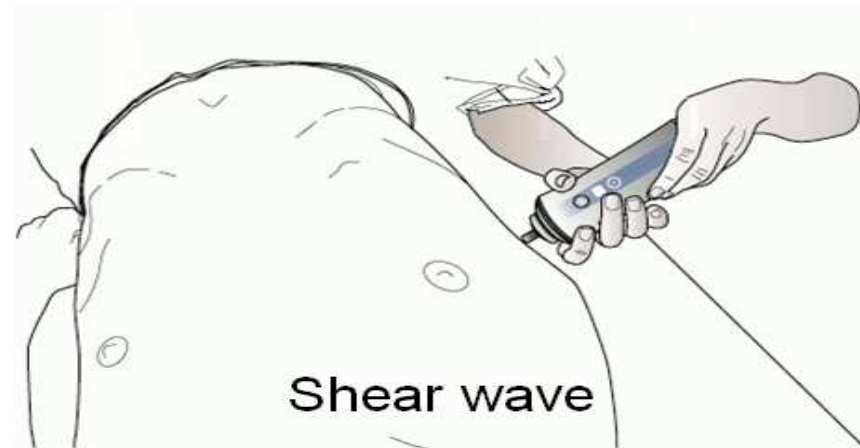


Cales P et al, Gastroenterol Clin Biol 2008; Liver Int 2008

“FibroScan” (Transient Elastography)



Transient elastography (TE) - Fibroscan



Sandrin L et al, Ultrasound Med Biol 2003;29:1705-13.

Transient elastography: principles

Normal liver (soft)



Cirrhosis (stiff)



Sandrin L et al, Ultrasound Med Biol 2003;29:1705-13.

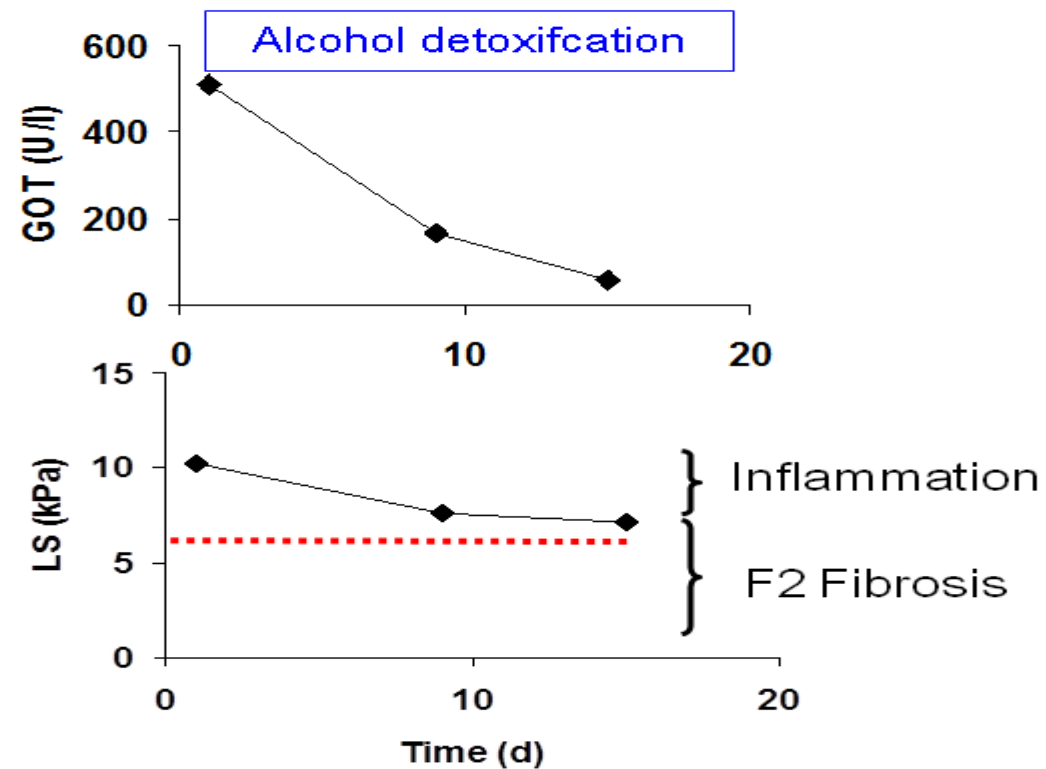
Factors affecting liver stiffness



Sagir, A. et al (2008) Hepatology 47, 592-595
Arena, U et al (2008) Hepatology 47, 380-384

Millonig G et al Hepatology 2008;48:1718-1723
Millonig G et al J Hepatol 2010;52:206-210.

LS rapidly decreases during alcohol detoxification



Mueller et al, World J Gastroenterol 2010 28; 16(8):966-972.

Man vs Machine:

Is FibroScan better than a hepatologist?

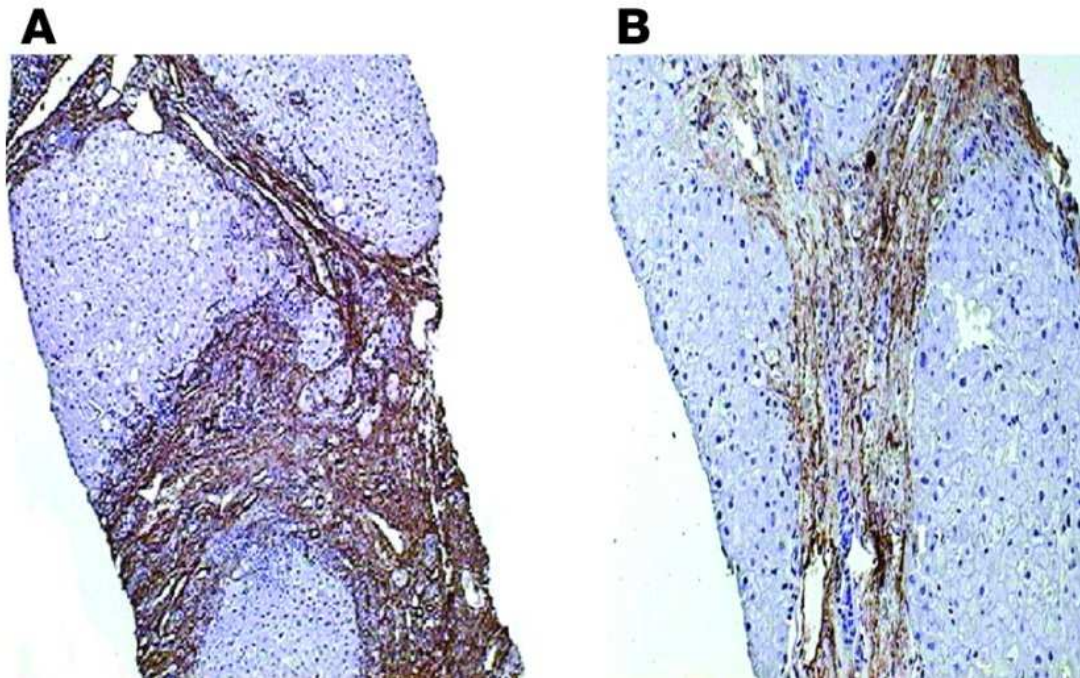
	Clinician	Clinical acumen AUC	Clinical acumen/lab tests/radiology AUC	TE alone AUC	p Value for AUC of TE vs clinical/lab/ radiology*
Detection of significant fibrosis	1	0.56	0.55	0.78	0.0003
	2	0.52	0.56	0.81	0.0001
	3	0.56	0.53	0.87	<0.0001
	4	0.59	0.52	0.78	<0.0001
Detection of cirrhosis	1	0.63	0.70	0.87	0.0268
	2	0.65	0.73	0.82	0.6693
	3	0.77	0.77	0.89	0.1367
	4	0.65	0.80	0.84	0.6705

Dolman GE et al. *Liver Int.* 2013

Reversibility of Fibrosis:

Liver biopsy from patient with hepatitis B infection

(A) Pre antiviral treatment, (B) post antiviral treatment



Kweon YO et al. *J Hepatol.* 2001

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Anti-fibrotic drugs in development

Agent	Main mechanism	Antifibrotic effects in HSCs	Antifibrotic effects in experimental fibrosis	Antifibrotic effect in humans
Angiotensin inhibitors	Inhibits HSC activation	Consistent positive data	Consistent positive data	Retrospective study
Colchicine	Inhibits inflammatory response	Limited data	Limited data	Discrepant results
Corticosteroids	Inhibits inflammatory response	Limited data	Limited data	Effective in autoimmune hepatitis
Endothelin inhibitors	Inhibits HSC function	Limited data	Limited data	Not tested
Interferon- α	Inhibits HSC activation	Consistent positive data	Consistent positive data	Effective in chronic hepatitis C
Interleukin 10	Inhibits inflammatory response	Limited data	Consistent positive data	Isolated reports in chronic hepatitis C
Pentoxifylline	Inhibits HSC activation	Consistent positive data	Consistent positive data	Not tested
Phosphatidylcholine	Decreases oxidative stress	Limited data	Consistent positive data	Not proven in alcohol-induced fibrosis
PPAR antagonists	Inhibits HSC activation	Consistent data	Consistent positive data	Isolated reports in NASH
S-adenosyl-methionine	Antioxidant	Limited data	Not tested	Effective in alcohol-induced fibrosis
<i>Sho-saiko-to</i>	Antioxidant	Consistent positive data	Consistent positive data	Isolated reports in chronic hepatitis C
TGF- β 1 inhibitors	Inhibits HSC activation and function	Consistent positive data	Consistent positive data	Not tested
Tocopherol	Antioxidant	Consistent positive data	Limited data	Isolated reports in NASH

Bataller R, Brenner DA. *J Clin Invest* 2005



Summary

- Chronic liver disease is a major problem in the UK – excess alcohol & obesity are a bad combination
- Most patients are asymptomatic until advanced fibrosis develops
- Fibrosis markers may be useful – but have some limitations
- We need better public health strategies



Portsmouth Hospitals **NHS**
NHS Trust

Dr Richard Aspinall
Consultant Hepatologist
Dept of Gastroenterology and Hepatology
Queen Alexandra Hospital
Portsmouth PO6 3LY

QAH
QUEEN ALEXANDRA
Hospital

richard.aspinall@porthosp.nhs.uk

liver@porthosp.nhs.uk

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